

# Yaohui Shu

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## SUMMARY

Ph.D. candidate in Applied Mathematics and Statistics working on causal inference for turbulent interactions in complex systems: high-dimensional, nonlinear, with feedback and unmeasured drivers, where correlation misleads. Builds minimal SDE models with known causal structure and exact results, then uses them as ground truth for information-theoretic and graph-based causal methods. First-author paper in *Geophysical Research Letters* on turbulent cloud physics.

## EDUCATION

**Stony Brook University**, Ph.D., Applied Mathematics & Statistics (GPA 3.78 / 4.0) Expected May 2027  
Dissertation: *Uncovering Causal Mechanisms of Aerosol-Cloud-Turbulence Interactions Using Physics- and Data-Driven Methods*  
Coursework: Advanced Stochastic Models & Risk Analysis, Numerical Analysis I–III, Financial Derivatives & Stochastic Calculus  
**University of Rochester**, M.S., Finance (GPA 3.81 / 4.0) Aug 2021 – Aug 2022  
Coursework: Derivatives Pricing, Financial Econometrics, Risk Management, Stochastic Calculus for Finance  
**CQF Institute**, Certificate in Quantitative Finance (CQF)

## SKILLS

|                            |                                                                                                                                                                     |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>         | Python (NumPy, SciPy, pandas, TensorFlow), MATLAB, C++, R; Git, SLURM/HPC                                                                                           |
| <b>Simulation</b>          | Explicit and implicit SDE integrators, Monte Carlo with time-step and sample-size convergence testing, constrained boundary handling, large parameter sweeps on HPC |
| <b>Stochastic Modeling</b> | Stochastic calculus, Fokker-Planck and Langevin equations, Ornstein-Uhlenbeck processes, maximum entropy, Weibull and gamma distribution families                   |
| <b>Statistics &amp; ML</b> | Causal discovery (FCI, PCMC, NOTEARS), kNN conditional mutual information, transfer entropy, doubly robust estimation (DML, AIPW, TMLE), LSTM, XGBoost, PCA         |
| <b>Finance</b>             | Derivatives pricing, financial econometrics, risk management, equity signal research, backtesting                                                                   |

## RESEARCH EXPERIENCE

**Stony Brook University** Stony Brook, NY  
*Doctoral Researcher: Stochastic Modeling, Monte Carlo & Causal Inference* Aug 2023 – Present

- Lead, as the group’s sole Ph.D. student, a research program on stochastic populations driven by a shared fluctuating environment under conservation laws, covering theory, simulation, causal analysis, and publication.
- Reconciled maximum-entropy statistics and Langevin dynamics for the same system: traced their disagreement to the choice of state variable and generalized the SDE model to any conserved moment order under one hypothesis.
- Identified a structural trade-off in finite-memory stochastic models (no boundary rule can conserve probability, preserve the driving process, and avoid mass at zero at once) and traced its effects in Monte Carlo results.
- Designing a causal discovery framework for high-dimensional, nonlinear spatiotemporal systems with feedback and unmeasured drivers (partial ancestral graphs, nearest-neighbor independence tests, path-space entropy rate as causal strength).
- Built stochastic benchmarks for how mixing with an external environment reshapes a population, spanning the continuum between two limiting regimes.

*Teaching Assistant*, AMS 595 Fundamentals of Computing (MATLAB, Python, C++) Aug 2026 – Present

**Brookhaven National Laboratory** Upton, NY  
*Research Assistant: Stochastic Processes, Monte Carlo & Information Theory* Jan 2024 – Jan 2026

- Showed where the white-noise diffusion approximation fails: extended it to SDEs with finite-memory (Ornstein-Uhlenbeck) noise, recovering the classical limit and quantifying how memory and intensity drive dispersion.
- Turned a solvable stochastic model into an exact benchmark for causal inference: derived time-resolved information flow in closed form and built an ensemble estimator that recovers causal direction where correlation cannot.
- Found that dispersion collapses onto the ratio of noise intensity to decorrelation rate, and that causal coupling between driver and population fades as the distribution spreads; published in *Geophysical Research Letters*.

**CQF Institute** Remote  
*Quantitative Finance Capstone: Equity Return Prediction* 2023

- Built an LSTM direction model from 156+ momentum, volatility, and technical features, cut to 10–15 via XGBoost importance, self-organizing maps, and PCA; leakage-free splits and a 162-configuration search.

## PUBLICATIONS & MANUSCRIPTS

- **Shu, Y.**, Liu, Y., Zhang, T., McGraw, R., Li, X. (2026). Stochastic Condensation with Information Flow for Causality Analysis. *Geophysical Research Letters*, 53(17), e2026GL123056. doi:10.1029/2026GL123056
- **Shu, Y.**, Liu, Y., Li, X. A Langevin-Type Stochastic Condensation Model to Explain the Size Distribution based on Maximum Entropy Principle. In preparation for *Physical Review E*.
- **Shu, Y.** Entropy-Based Causal Discovery for Spatiotemporal Dynamical Systems with Feedback and Unmeasured Drivers. Methodology working paper; one further manuscript in preparation.
- Conference abstracts: AGU Fall Meeting 2025 (A21B-2034, first author; A21B-203); 106th AMS Annual Meeting 2026.